



Modeling grazing effects on carbon dynamics in alpine grasslands of the Qinghai-Tibetan Plateau using the Biome-BGCMuSo model

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ABSTRACT

The alpine grassland ecosystem on the Qinghai-Tibetan Plateau (QTP), the world's most elevated pastoral region, is highly sensitive to climate change and disturbance caused by grazing. Due to the extensive nomadic grazing system on the QTP, incorporating seasonality into livestock spatial distribution is essential for understanding the complex system interactions between climate, vegetation, and grazing. Here, the process-based biogeochemical model Biome-BGCMuSo was enhanced by incorporating a winter grazing module to simulate grassland carbon dynamics on the QTP. The particle swarm optimization algorithm was employed to optimize the most sensitive parameters of Biome-BGCMuSo screened out by Sobol' method. The simulation performance was significantly improved by optimizing model parameters. Based on regional simulations, the alpine grassland ecosystem on the QTP acted as a carbon sink of 8.72 Tg C a⁻¹ (1 Tg = 10¹² g) during 1979–2018 and 16.31 Tg C a⁻¹ during 2002–2018. From 1979–2018, total carbon increased by 0.34 Pg C (1 Pg = 10¹⁵ g), primarily due to an increase in soil organic carbon (SOC), with 77.5 % of the carbon being sequestered above the soil depth of 30 cm. In grasslands on the QTP, model revealed that grazing reduced vegetation carbon, litter carbon, and top layer (0–30 cm) soil organic carbon by 27.8 g C m⁻², 17.0 g C m⁻², and 270.1 g C m⁻², respectively, indicating that grazing had detrimental effects on regional carbon storage capability. The model implied that summer grazing had a greater effect on carbon stocks than winter grazing at the same grazing intensity, especially in SOC. Our study recommends that summer grazing and winter grazing should be considered separately to understand the complex relationships between grazing management and ecosystem carbon dynamics, which could provide a new perspective for setting grazing safety thresholds to ensure sustainable development of grasslands.

1. Introduction

The Qinghai-Tibetan Plateau plays a significant role in the global carbon (C) cycle due to its high carbon density and large area (Chen et al., 2020). The Qinghai-Tibetan Plateau has an average elevation of over 4000 m, with an area of approximately 2.5 × 10⁶ km² (Bibi et al., 2018). The soil organic carbon (SOC) of the Qinghai-Tibetan Plateau was estimated at 36.6 Pg C (1 Pg = 10¹⁵ g), and the ecosystems on the Qinghai-Tibetan Plateau acted as a carbon sink for 26.5–33.7 Tg C a⁻¹ (1

Tg = 10¹² g), representing 9.9–19.6 % of the terrestrial carbon sink in China (Ding et al., 2019; Wang and Piao, 2023). More than 60 % of the Qinghai-Tibetan Plateau is covered by grasslands, which are the Earth's largest alpine ecosystem (defined as above timberline), shaped by the complexity of human interaction with these high-elevation environments (Xia et al., 2018; Miede et al., 2014). The grasslands of the Qinghai-Tibetan Plateau have served Tibetan communities as a predominant pasture for thousands of years, making it one of the most important pastoral areas in China (Huang et al., 2017; Lehnert et al.,

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2016). There is considerable controversy regarding the extent to which grazing contributes to alpine grasslands dynamics on the Qinghai-Tibetan Plateau (Li et al., 2018). Specifically, livestock feeding, herbivore trampling, and returning excreta are acknowledged factors affecting grassland vegetation and SOC, which are essential to the evaluation of the ecological effects of livestock grazing. The impact of grazing on vegetation carbon (VEGC) and SOC varies with grazing intensity and climate conditions, thereby increasing the complexity of ecological processes in alpine grassland ecosystems (Liu et al., 2020; McSherry and Ritchie, 2013). In general, grazing reduces aboveground biomass (AGB); however, in certain instances, light grazing may result in an increase in AGB due to overcompensation growth effects (Li et al., 2017; Luo et al., 2012). There are different effects of grazing intensities on SOC; heavy grazing may significantly reduce SOC, whereas light grazing may increase SOC, possibly due to grazing-induced biomass transfer (Zhan et al., 2020). While AGB may be reduced due to light grazing, belowground biomass can increase, thereby facilitating organic matter input into the soil and increasing SOC.

On the Qinghai-Tibetan Plateau, the alpine grasslands are typically divided into winter pastures (also known as winter-spring pastures and cold-season pastures) and summer pastures (also known as summer-autumn pastures and warm-season pastures) (Huang et al., 2017; Zhan et al., 2023). Traditionally, seasonal rotational grazing is characterized by the movement of livestock between summer and winter pastures. This enables herders to utilize environmental resources in a manner that is consistent with the seasonal fluctuations in climate and grassland productivity. Summer pastures are generally located in high mountainous and subalpine regions with high terrain, in areas where vegetation grows abundantly during the summer (Huang et al., 2017; Wang et al., 2010). Winter pastures are generally located near residential areas, typically in low-elevation areas, such as plains and low mountains, where low terrain and early spring returns facilitate grass rejuvenation in the spring (Fan et al., 2010; Wang et al., 2010). Summer pasture is used by herders from June to October for nomadic grazing while living in tents, and winter pasture is used from November to May for grazing while living in permanent homesteads (Huang et al., 2017). It is customary for livestock to be moved between high-elevation pastures during the summer and low-elevation pastures during the winter by their herders. It is of paramount importance to comprehend the disparate impacts of winter grazing (cold-season grazing) and summer grazing (warm-season grazing) on alpine grasslands. A multisite survey revealed that plant biomass, soil moisture, and the storage of soil carbon and nitrogen were significantly higher in winter pastures compared to summer pastures in alpine grassland ecosystems on the Qinghai-Tibetan Plateau (Tian et al., 2020).

Process-based models are valuable tools for assessing a wide range of feedbacks within the terrestrial biosphere and the impacts of human activities on it (Schaphoff et al., 2018). The importance of grazing in grassland ecosystems has led many process-based models to incorporate grazing processes. For example, Vuichard et al. (2007) included a grazing module in the PaSim model based on grassland utilization in Europe. The grazing module of PaSim has been integrated into ORCHIDEE (Chang et al., 2013) and LPJmL (Rolinski et al., 2018). It was used to estimate the minimum managed grassland area within 0.5°, proportions of grazing and mowing grassland, and the grazing intensity on a global scale (Chang et al., 2016). In grassland ecosystems with little or no mowing, the defoliation equation by Seligman et al. (1992) is a pivotal tool for estimating livestock consumption of AGB based on grazing intensity. Process-based models that incorporate this equation and utilize grazing intensity as input data are capable of quantification of herbaceous biomass removal caused by livestock grazing. The defoliation equation by Seligman et al. (1992) has been incorporated into various process-based biogeochemical models to assess the impact of grazing across different regions. Specifically, the Sim-CYCLE model (Chen et al., 2007) and the DLEM model (Dangal et al., 2016) have been applied to the grasslands of the Mongolian Plateau. Furthermore, the

Biome-BGC model (Han et al., 2016; Luo et al., 2012), and the AEM model (Zhu et al., 2022) have been applied to the grasslands of Central Asia. These models have been instrumental in assessing the effect of grazing on the carbon cycle, evapotranspiration, and water use efficiency in their applied regions. Although many process-based models have grazing modules, their application to the Qinghai-Tibetan Plateau remains challenging since previous grazing studies using process-based models focused on summer grazing and were unable to simulate winter grazing (Huang et al., 2022; Zhang et al., 2017). During the winter grazing period, livestock consume withered grasses that have accumulated in the wild. These models do not include winter grazing due to a lack of the carbon and nitrogen pools of dead standing biomass, or because the dead standing biomass pools are not available for livestock to forage. Moreover, the lack of spatial data pertaining to pasture distribution and grazing intensity during the summer and winter seasons adds to the difficulty of modelling the winter grazing.

The process-based biogeochemical model Biome-BGC has been extensively applied to the Qinghai-Tibetan Plateau with the purpose of simulating gross primary productivity (GPP) and net primary productivity (NPP) and of evaluating the impact of temperature and water stress on ecosystem productivity (Sun et al., 2017; Ye et al., 2013; You et al., 2020). Biome-BGC exhibits certain structural deficiencies, including an oversimplified soil moisture module relating to only one soil layer, the accuracy of phenology, and the deficiency in simulation of grassland ecosystem responses to drought stress (Hidy et al., 2016). Biome-BGCMuSo (where MuSo represents a multilayer soil module) is a new version of Biome-BGC that incorporates several structural modifications and new developments in vegetation management modules to enhance its suitability for simulating vegetation dynamics and carbon dynamics in grassland ecosystems (Hidy et al., 2022; Hidy et al., 2016). Biome-BGCMuSo is still open-source code, which allows new modules to be added. In our previous work, we calibrated three phenological models for the start of the growing season and five phenological models for the end of the growing season using remote sensing data and selected the best phenological models for Biome-BGCMuSo on the Qinghai-Tibetan Plateau (Zheng et al., 2022). This study aims to further develop and improve the grazing module of Biome-BGCMuSo according to the actual utilization of grasslands on the Qinghai-Tibetan Plateau.

Therefore, this study used the Biome-BGCMuSo model to assess the ecological impacts of grazing on grasslands on the Qinghai-Tibetan Plateau under climate change. We aimed to (1) incorporate the defoliation equation of Seligman et al. (1992) and winter grazing process into the Biome-BGCMuSo model to improve grazing simulations, (2) provide an in-depth investigation into the vegetation and carbon dynamics of the grasslands on the Qinghai-Tibetan Plateau, with a particular focus on quantifying the impacts of grazing on vegetation and carbon dynamics.

2. Materials and methods

2.1. Study area

Qinghai-Tibetan Plateau is located in southwestern China, between latitudes 25.9°N and 39.8°N, and longitudes 73.4°E and 104.8°E. The average annual temperature for the period 1982–2018 ranged from −10°C in the northern region to 23°C in the southern region, with 67 % of the Qinghai-Tibetan Plateau experiencing an annual temperature below 0°C (Peng et al., 2021). Precipitation occurs mainly in summer, with mean annual precipitation increasing southeastward from less than 100 mm to more than 700 mm, with 54 % of the region receiving precipitation below 400 mm a^{−1} during 1982–2018 (Peng et al., 2021; Yang et al., 2010). The annual net short-wave radiation ranges from 2197 W m^{−2} to 3755 W m^{−2} during 1982–2018 (Peng et al., 2021). This is due to the high altitude, low latitude, and cloudless sky. Along the precipitation gradient, two principal types of grasslands can be identified: alpine steppe, which is characterized by drought-tolerant graminoid species, and alpine meadow, which is dominated by mesic and

meso-xeric species with a characteristic root mat (Wang et al., 2022) (Fig. 1a).

2.2. Model description

Biome-BGCMuSo (version 6.1), a modified version of Biome-BGC, was used in this study. Biome-BGCMuSo simulates the storage and fluxes of carbon, nitrogen, and water within and between the components of terrestrial ecosystems, including vegetation, litter, and soil (Fig. 2). Using a daily time step, the model is driven by daily values of precipitation, solar radiation, maximum and minimum temperatures, and vapor pressure deficit (VPD). Biome-BGCMuSo is structurally improved compared to Biome-BGC (including replacing the single layer soil module with a multilayer soil module, improving the phenological module, and developing a soil moisture-related plant senescence module); in addition, there are management modules that simulate cropland management (planting, fertilization, harvesting), forest management (thinning), and grassland management (mowing, grazing) (Hidy et al., 2016). The incorporation of a ten-layer soil module into Biome-BGCMuSo has enabled the model to achieve a more sophisticated representation of water and carbon dynamics. This has enhanced the simulation of soil-water-related processes, carbon, and nitrogen cycling through distinct soil layers (Hidy et al., 2022). The grazing module is incorporated within Biome-BGCMuSo, and more detailed information about the grazing module can be found in Hidy et al. (2012). The spin-up simulations of the Biome-BGCMuSo were run for a maximum of 3200 years, and the transient simulations for 240 years in this study.

2.3. Improvement of grazing module and transient run

2.3.1. Estimation of the grass consumption rate

The original grazing module of the Biome-BGCMuSo model simulates the daily consumption of both leaf and soft stem (soft stem is non-photosynthesizing, aboveground, non-woody biomass in the model) biomass by animals (Hidy et al., 2012). Grazing is defined based on the “livestock unit” (LSU) terminology, where a LSU represents an average animal (the exact animal-specific value can be set). In our study, 1 LSU represents the equivalent of one standard sheep unit. The impact of grazing on carbon in the model is determined by the carbon loss caused by grazing (C_{demand}), which is a function of grazing intensity:

$$C_{demand} = G_I \times D_X \quad (1)$$

where C_{demand} , G_I and D_X represent the herd's demand ($\text{g C ha}^{-1} \text{ day}^{-1}$), grazing intensity (LSU ha^{-1}), and satiation consumption rate of the livestock ($\text{g C day}^{-1} \text{ LSU}^{-1}$), respectively.

The original model does not take into account leaves and soft stems that cannot be consumed by animals as well as the area of land each

animal can cover daily. Hence, the defoliation equation developed by Seligman et al. (1992) and the revision of the grazing module of Zhu et al. (2022) were incorporated into the Biome-BGCMuSo. Leaf and soft stem consumption by the animals (C_{graze} , $\text{g C ha}^{-1} \text{ day}^{-1}$) was determined by the balance between the herd's demand (C_{demand}) and the amount of leaves and soft stems available to the animals (C_{supply} , $\text{g C ha}^{-1} \text{ day}^{-1}$):

$$C_{graze} = \min(C_{demand}, C_{supply}) \quad (2)$$

$$C_{supply} = \text{Area}_{graze} \times C_{leaf+softstem,av} \quad (3)$$

$$\text{Area}_{graze} = G_e \times G_I \quad (4)$$

$$C_{leaf+softstem,av} = C_{leaf+softstem} - C_{leaf+softstem,r} \quad (5)$$

where Area_{graze} is the daily grazed grassland area (ha); $C_{leaf+softstem,av}$ is the grass available to livestock (g C m^{-2}); G_e is the mean area of land that can be covered by a LSU each day ($\text{ha day}^{-1} \text{ LSU}^{-1}$); $C_{leaf+softstem}$ is the total leaf and soft stem carbon of the grassland (g C m^{-2}); $C_{leaf+softstem,r}$ is the leaf and soft stem remains that cannot be consumed by LSU (g C m^{-2}).

2.3.2. Winter grazing module implementation

Winter grazing on the Qinghai-Tibetan Plateau is not currently considered in Biome-BGCMuSo. This is because livestock in winter pastures are primarily fed standing dead grasses accumulated during the growing season (since winter pastures are not grazed during the summer). The original grazing module of Biome-BGCMuSo only provides live leaf carbon and soft stem carbon to livestock. Since the grassland plants in the model are annual, there will be no live leaves or soft stems on the ground during the winter, preventing the winter grazing process. The winter grazing process was achieved here using the standing dead biomass (STDB) pool, which is an intermediate carbon pool between the plant aboveground carbon pool and the litter carbon pool (Fig. 1b). When plants are wilting, their leaf biomass and soft stem biomass is transferred directly to the litter carbon pool, as described in the original model. However, in this study, the withered parts of leaf and soft stem biomass were initially allocated to the STDB pool before being transferred to the litter carbon pool. STDB can be fed to livestock in winter when leaf biomass and soft stem biomass are not available. STDB and living AGB (leaf biomass and soft stem biomass) are often present simultaneously at the beginning and end of the growing season. Therefore, livestock feeding requirements were allocated to both the live AGB pool and the STDB pool based on the ratio between the existing STDB and the live AGB. There were some differences in the grazing module parameters between the winter pasture and the summer pasture, which is attributable to livestock movement throughout the season, and

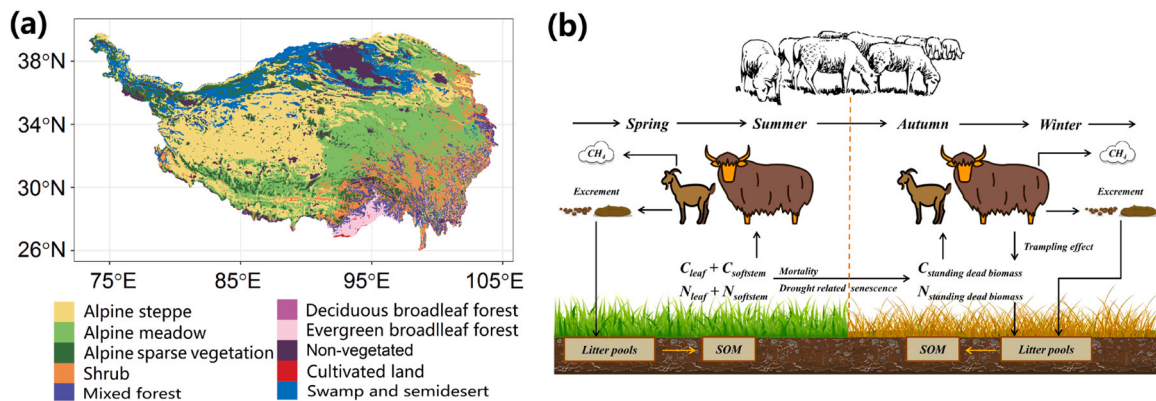


Fig. 1. Land cover types on the Qinghai-Tibetan Plateau, extracted from the Vegetation Maps of China on a scale of 1:1000000 (a). Illustration of the grazing module of the Biome-BGCMuSo (b).

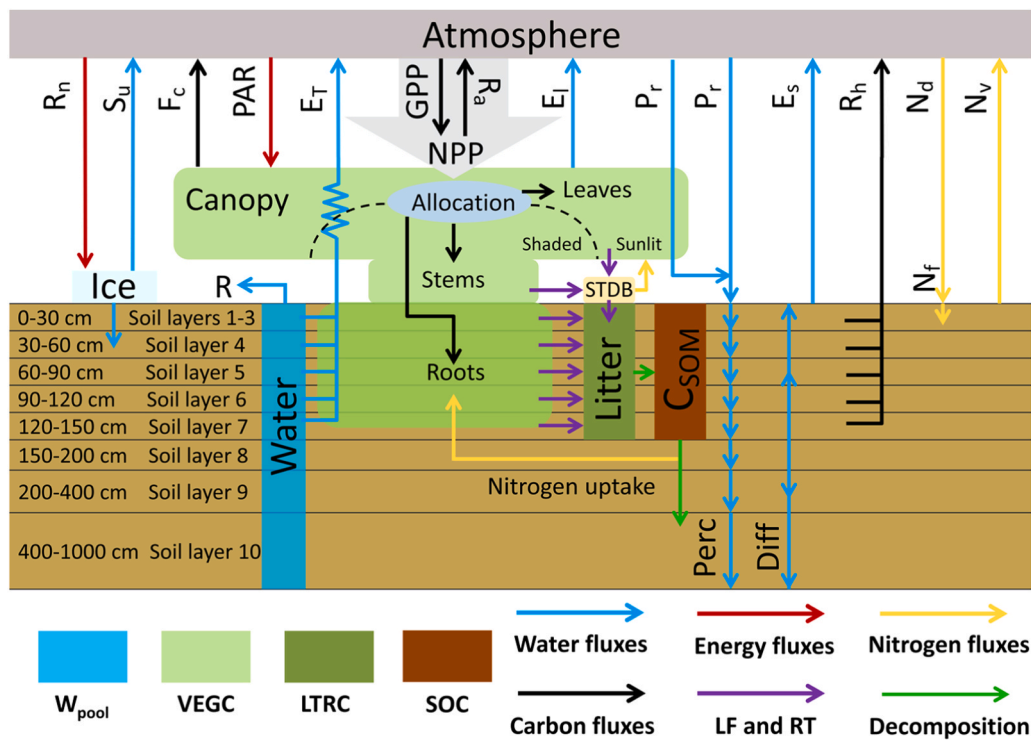


Fig. 2. Biome-BGCMuSo framework for representing the fluxes of carbon, water, energy, and nitrogen within the model. R_n : incident radiation (canopy transmitted radiation); PAR: photosynthetically active radiation; E_i : interception; E_t : transpiration; E_s : evaporation; Perc: percolation; Diff: diffusion; P_r : precipitation; GPP: gross primary productivity; NPP: net primary productivity; R_a : autotrophic respiration; R_h : heterotrophic respiration; F_c : carbon emitted by fire; R : runoff; S_u : sublimation; N_d : nitrogen deposition; N_f : nitrogen fixation; N_v : nitrogen volatilization; STDB: standing dead biomass; SOM: soil organic matter; LF: litterfall; RT: root turnover; W_{pool} : water pool; VEGC: vegetation carbon; LTRC: litter carbon; SOC: soil organic carbon. Litterfall, root turnover, and decomposition constitute essential carbon fluxes in the ecosystem carbon cycle. These processes are depicted with distinct colors for visual emphasis. STDB is a component of VEGC.

the physiological changes observed in the livestock during this time period (Table S1).

2.3.3. Improvement of the grazing-related nitrogen cycle

In original Biome-BGCMuSo, only feces were considered for excreta, with the nitrogen content of feces set at a fixed value (i.e., parameter nitrogen content of excrement). This study assumed that 80 % of the nitrogen consumed by livestock was returned to the environment through livestock excreta (Baron et al., 2002). Particularly, this study distinguished nitrogen from urine and feces since urinary nitrogen constitutes 65 % of the nitrogen in excreta (Baron et al., 2002) and is readily absorbed by plants (Li, 2021). The nitrogen in the urine is transferred into the rapid turnover pool of the litter nitrogen pool.

2.3.4. Improvement of the transient run

In Biome-BGCMuSo, the spin-up simulation (equilibrium run) starts with low initial soil carbon and nitrogen levels and progresses until a steady state is reached, simulating pre-industrial conditions without altering CO_2 concentrations or nitrogen deposition. Inconsistent CO_2 and nitrogen deposition data between the spin-up simulation and normal simulation (representing current conditions) may result in undesirable transient behavior of the model. To minimize undesired transient behavior caused by abrupt environmental shifts between the spin-up simulation and the normal simulation, the transient simulation was implemented in Biome-BGCMuSo (Hidy et al., 2016). The original Biome-BGCMuSo model only reused meteorological data once, effectively repeating a normal run. We enhanced the transient run by enabling the user to specify the simulation duration via the initialization file. This modification facilitates a smoother transition by gradually adjusting key environmental drivers. Specifically, CO_2 concentration and nitrogen deposition gradually increase from the industrial revolution value to the level recorded in the year preceding the normal

simulation, while grazing intensity increases linearly from 0 $LSU\ ha^{-1}$ to grazing intensity in the first year of the normal simulation.

2.4. Scenarios design

In order to quantify the impact of grazing on alpine grassland carbon dynamics, we designed two experimental scenarios (Table S2). $S_{GRAZING}$ scenario (real scenario) simulates actual ecosystem conditions based on historical meteorological data (1979–2018), observed atmospheric CO_2 concentrations (1979–2018), and spatially explicit grazing intensity data (1979–2018). This scenario captures carbon fluxes under the concurrent influences of climate change, CO_2 fertilization, and grazing pressure in alpine grasslands. $S_{NON-GRAZING}$ scenario represents a counterfactual experiment in which grazing is excluded, while meteorological data and CO_2 concentrations remain the same identical to those in the $S_{GRAZING}$ Scenario. This design allows for the isolation of grazing effects through differential comparison between two scenarios.

2.5. Data collection and processing

2.5.1. Model input data (meteorological and environmental data)

Regional daily meteorological data, comprising precipitation ($mm\ day^{-1}$), shortwave radiation ($W\ m^{-2}$), and daily minimum, maximum, and mean temperatures ($^{\circ}C$), with a spatial resolution of 0.1° , was obtained from the China Meteorological Forcing Dataset for the period 1979–2018 (He et al., 2020). The MTCLIM model was used to calculate the daily VPD (Pa) and photoperiod data (s) (Bohn et al., 2013). Based on the 1:1 million Vegetation Map of China, the area proportion of alpine meadows and steppes was calculated at a spatial resolution of 0.1° (Fig. S1). The data on soil texture, including percentages of sand, silt, and clay, as well as pH values were obtained from the China soil characteristics dataset (2010) (Shangguan et al., 2012). The Vegetation

Map of China and the China soil characteristics dataset (2010) were obtained from the National Tibetan Plateau Data Center (<http://data.tpdc.ac.cn/>). The elevation data were obtained from the Digital Elevation Model (DEM) of the Shuttle Radar Topography Mission (SRTM) v4.1 data. The shortwave albedo of bare soil (albedo is LAI-dependent in Biome-BGCMuSo) was generated using the MODIS daily albedo product (MCD43C3, Collection 6). The spatial pattern of the albedo of bare soil on the Qinghai-Tibetan Plateau can be found in Fig. S2, and more information on the method used to generate bare soil shortwave albedo can be found in Supplementary Text S1.

2.5.2. Model input data (grazing data)

In our study, the cold season, which extends from November to May, and the warm season, which spans from June to October, delineate the traditional Tibetan transhumance pattern. The impact of grazing on grassland vegetation was assessed using this seasonal division. It is evident that grazing affects grassland vegetation differently in the cold and warm seasons. During the summer, livestock consume fresh grass, whereas during the winter, they consume dead grass in the field (dead standing biomass). This study used elevation data as a key determinant of pasture division for cold and warm seasons. We used the elevation division criteria proposed by Wang et al. (2010) for seasonal pastures in eight prefecture-level divisions in Qinghai Province (hereafter referred to as Qinghai). In the six prefecture-level divisions in the Tibetan Autonomous Region (hereafter referred to as Tibet), we dynamically adjusted the elevation criteria for pasture division in the cold and warm seasons to ensure that the final distribution of seasonal pastures is consistent with the pasture area ratio in the cold and warm seasons as determined by The comprehensive scientific expedition team of the Qinghai Tibet Plateau, Chinese Academy of Sciences (1992). Distribution of pastures in the cold and warm seasons in Xinjiang Autonomous Region (hereafter referred to as Xinjiang) was obtained from Zhu et al. (2022). Since there are no elevation division criteria for Sichuan (hereafter referred to as Sichuan), Yunnan Province (hereafter referred to as Yunnan), and Gansu province (hereafter referred to as Gansu), the nearby regions of Tibet and Qinghai were used as a reference for determining pasture distribution in the cold and warm seasons. The distribution of summer pasture and winter pasture on the Qinghai-Tibetan Plateau was mapped using area ratios of different pasture types in alpine meadows or alpine steppes within the 0.1° grid (Fig. S3).

The grazing intensity data were extracted from the Gridded Livestock of the World (GLW3) database, which reflects the subnational livestock distribution data in 2010 (Gilbert et al., 2018). The GLW3 was generated using a random forest (RF) algorithm and provides global population densities of cattle, buffaloes, horses, sheep, goats, pigs, chickens, and ducks in each land pixel at a spatial resolution of 0.083° . The livestock numbers from the statistical yearbooks of Tibet, Qinghai, Xinjiang, Gansu, Sichuan and Yunnan, as well as sequence data of livestock number at county level on the Qinghai-Tibetan Plateau and the data on livestock products from the Tibetan Autonomous Region obtained from the National Tibetan Plateau Data Center (<http://data.tpdc.ac.cn/>), were interpolated and converted year by year based on GLW3 data in order to determine the grazing intensity from 1979 to 2018. The Supplementary Text S2 and Figs. S4–S6 presents more detailed information about the process of generating grazing intensity data.

2.5.3. Calibration and validation data for AGB and SOC

We collected field-observed AGB and SOC in fenced plots and in controlled experiments with no-grazing treatments from previous publications on the TP to calibrate and validate the Biome-BGCMuSo in $S_{\text{NON-GRAZING}}$ scenario. When data were presented only in graphical form, we used the GetData Graph Digitizer (version 2.25) to extract numerical information. In total, 191 plots of AGB and 87 plots of SOC were used for calibration and validation (Table S3, S4). The field-observed AGB and SOC datasets were each randomly divided into two

parts for calibration and validation, respectively. For calibration, 80 % of the data was used, while the remaining 20 % was employed for validation. The field-observed data were compared with the AGB and SOC extracted from the regional simulation in the same year. More detailed information about the field-observed AGB and SOC were provided in Supplementary Text S3 and Fig. S7. It is important to note that these observational data mainly represent spatial gradients of environmental factors across different sites (e.g., variations in precipitation, temperature, elevation, etc.), rather than temporal changes in AGB and SOC within a single site over a long period of time. We obtained regional scale AGB data from Zeng et al. (2019) to validate the Biome-BGCMuSo in S_{GRAZING} scenario. The data were monitored by a RF algorithm, using data from 256 field-observed AGB, remotely sensed vegetation indexes, meteorological information, and topographical information. Spatial AGB data were high-resolution (1×1 km) and aggregated to $0.1^\circ \times 0.1^\circ$ resolution for model use. The AGB data required for the validation of the grazing module of the Biome-BGCMuSo model were obtained from Ren et al. (2021) by using GetData Graph Digitizer software. The grazing control experimental platform of Ren et al. (2021) is situated at $102^\circ 33'E$, $32^\circ 48'N$ on the Qinghai-Tibetan Plateau Research Base of Southwest Minzu University. Ren et al. (2021) incorporated four grazing intensity treatments, designated as 0, 1, 2, and 3 yak/ha, corresponding to 0, 4.5, 9.0, and 13.5 livestock units (LSU)/ha, based on the standard conversion of 1 yak = 4.5 LSU according to the national standard of China (NY/T 635–2002). Grazing commenced in late May and concluded in late September annually from 2015 to 2018. In late August of 2018, AGB was collected from plots.

2.6. Model calibration and validation

2.6.1. Sensitivity analysis

Considering the complexity of the Biome-BGCMuSo model, which contains numerous parameters, optimizing these parameters is essential, yet challenging (Table S5). To address this issue, a sensitivity analysis was conducted in order to identify key parameters that influence the simulation variables. This study focused on the ecophysiological parameters of alpine meadows and steppes, which are considered functional types of C3 grasses. Employing the Sobol' method, a quantitative method of global sensitivity analysis, allowed us to decompose the output variance into the contributions attributable to each input factor (Sobol, 1993). To reduce the computational requirements, we performed a One-At-a-Time (OAT) sensitivity analysis which removed all parameters that had no effect on AGB or SOC. Furthermore, no periodic fires have been observed in alpine grasslands of the Qinghai-Tibetan Plateau in recent decades, so we set the annual fire mortality parameter to zero (You et al., 2019). Finally, the parameters involved in the Sobol' sensitivity analysis were determined. These parameters fall under the categories of plant functioning parameters and phenological (allocation) parameters of the ecophysiological input file. The probability distribution functions (PDFs) of the parameters were obtained from relevant literature (Table 1). The R package "sensitivity" was used to conduct a sensitivity analysis based on the Sobol' method. The "sobolmartinez" algorithm from the R package "sensitivity" was used in this study, which implements the Monte Carlo estimation of the Sobol' indices for both first-order and total indices using correlation coefficients-based formulas, at a total cost of $(p + 2) \times n$ model evaluations, where p represents the number of input parameters (28 in this study) and n represents the number of base samples (set to 3000 in this study). We considered parameters with a total-order sensitivity index greater than 0.1 as sensitive parameters. There are some differences between alpine meadows and alpine steppes in terms of physiological characteristics (Yang et al., 2009a). To more accurately quantify the carbon dynamics of alpine grasslands on the Qinghai-Tibetan Plateau, we conducted sensitivity analysis and parameter optimization separately on alpine meadows and alpine steppes (You et al., 2019).

Table 1

Distribution function and range of ecophysiological parameters of alpine grasslands.

No.	Parameters	pdf	Distribution	Source
1	TGGS	Uniform	U(0.18, 0.54)	Engstrom et al. (2006); Wang (2014)
2	LGS	Uniform	U(0.135, 0.405)	Sun et al. (2017)
3	WPM	Uniform	U(0.05, 0.15)	Default*
4	C:N _{leaf}	Normal	N(25, 8.6)	White et al. (2000)
5	C:N _{lit}	Normal	N(45, 11)	White et al. (2000)
6	C:N _{fr}	Normal	N(50, 19)	White et al. (2000)
7	C:N _{ss}	Normal	N(25, 8.6)	Set same with C:N _{leaf}
8	L _{lab}	Normal	N(0.68, 0.086)	Calculated from White et al. (2000)
9	L _{cel}	Normal	N(0.23, 0.077)	White et al. (2000)
10	FR _{lab}	Uniform	U(0.144, 0.527)	Calculated from White et al. (2000)
11	FR _{cel}	Uniform	U(0.378, 0.495)	White et al. (2000)
12	SS _{lab}	Normal	N(0.68, 0.086)	Set same with L _{lab}
13	SS _{cel}	Normal	N(0.23, 0.077)	Set same with L _{cel}
14	W _{int}	Uniform	U(0.018, 0.032)	White et al. (2000)
15	LEC	Normal	N(0.48, 0.13)	White et al. (2000)
16	FLNR	Uniform	U(0.075, 0.225)	Default*
17	G _{smax}	Uniform	U(0.0025, 0.0075)	Default*
18	G _{bl}	Uniform	U(0.02, 0.06)	Default*
19	MDRZ	Uniform	U(0.5, 1)*Soil Depth	Set 50 %–100 % of soil depth
20	RDP	Uniform	U(5, 10)	Calculated from Li et al. (2011)
21	RWMRD	Uniform	U(0.8, 3.5)	Calculated from Li et al. (2011)
22	RDFSP	Uniform	U(0.05, 0.5)	Calculated from Li et al. (2011)
23	NSC _{mr}	Uniform	U(0.15, 0.45)	Default*
24	N _{fixation}	Uniform	U(0.00075, 0.000225)	Zhu (1983)
25	TD	Uniform	U(1, 3)	Dyukarev (2017)
26	LA	Uniform	1-FRA-SSA	/
27	FRA	Uniform	U(0.38, 0.87)	Zhang (2018)
28	SSA	Uniform	U(0.016, 0.280)	Wang et al. (2017)

* The parameters is assumed to be uniformly distributed and its variation range is obtained by varying its default by $\pm 50\%$; part of the parameters distributions is described in You et al. (2019); Table S5 presents the description of each parameter.

2.6.2. Parameter optimization

The particle swarm optimization (PSO) algorithm, which is inspired by the behavior of swarms of animals, is a very popular stochastic population-based algorithm (Eberhart and Kennedy, 1995). We implemented the R package “hydroPSO” for optimizing parameters that had been identified as significant by Sobol’ sensitivity analysis method. We used the “spso2011” algorithm in the R package “hydroPSO” with a set of arguments ($q = 30$, $N = 30$, i.e., 30 particles; 30 iterations). The weighted average Kling-Gupta efficiency (KGE) (Gupta et al., 2009) for values of AGB and SOC were used as the objective function in the PSO algorithm. KGE is defined as follows:

$$KGE = 1 - \sqrt{(r-1)^2 + (\alpha-1)^2 + (k-1)^2} \quad (6)$$

where r , α , and k represent the correlation coefficient, the standard deviation ratio and the mean value ratio between the observed and simulated values, respectively. The KGE value range is $[-\infty, 1]$, with values closer to 1 indicating better results. According to the PSO algorithm, the optimal parameter values are shown in Table S6.

2.6.3. Statistical analysis

In order to evaluate the performance of the model simulations, two commonly used evaluation measures are employed: the coefficient of determination (R^2) and the root mean square error (RMSE). These are defined below:

$$R^2 = \frac{\sum_{i=1}^n (S_i - \bar{O})^2}{\sum_{i=1}^n (O_i - \bar{O})^2} \quad (7)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (O_i - S_i)^2}{n}} \quad (8)$$

where S_i and O_i represent the i th simulated values of AGB and SOC from the Biome-BGCMuSo simulations and the corresponding field-observed values, respectively. For regional-scale validation, S_i represents the i th simulated AGB value from the Biome-BGCMuSo simulations, while O_i represents the corresponding AGB value monitored by a RF algorithm, serving as the observed data.

3. Results

3.1. Sensitivity analysis, model calibration and validation

In general, the four most sensitive parameters for both AGB and SOC were C:N of leaves (C:N_{leaf}), fine root allocation (FRA), soft stem allocation (SSA), and root depth function shape parameter (RDFSP) (Fig. 3). C:N_{leaf} and fine root allocation were found to have the greatest impact on SOC and AGB simulations, respectively. In alpine steppes, the total order sensitivity of the C:N_{leaf} was approximately twice that of the first order (Fig. 3a, c), suggesting that direct and indirect effects of the C:N_{leaf} have approximately the same impact on the simulation. It can be observed that the fine root allocation and soft stem allocation parameter affect SOC simulation mainly through interaction, as evidenced by the small first-order sensitivity index values (Fig. 3c, d). The parameters that had significant effects on AGB and SOC were not significantly different between alpine meadows and alpine steppes. Based on a threshold of 0.1 for the total-order sensitivity index, we optimized the following parameters: C:N_{leaf}, C:N of leaf litter, after retranslocation (C:N_{lit}), C:N of fine roots (C:N_{fr}), leaf litter labile proportion (L_{lab}), leaf litter cellulose proportion (L_{cel}), soft stem litter cellulose proportion (SS_{cel}), fraction of leaf N in Rubisco (FLNR), maximum stomatal conductance (G_{smax}), maximum depth of rooting zone (MDRZ), root distribution parameter (RDP), root depth function shape parameter, fine root allocation, soft stem allocation.

Parameter optimization substantially improved model performance, achieving maximum KGE values of 0.55 for alpine meadows (Fig. 4a) and 0.63 for alpine steppes (Fig. 4b) in the 28th iteration of the PSO algorithm. The calibrated Biome-BGCMuSo simulated well the AGB and SOC compared to observations, with coefficients of determination (R^2) of 0.75 and 0.58, respectively (Fig. 5a1, a2). The validation of the Biome-BGCMuSo model resulted in R^2 values of 0.62 and 0.67 for AGB and SOC, respectively, when comparing simulations to observations (Fig. 5b1, b2). The model had a tendency to overestimate SOC, especially at high values (Fig. 5a2). The Biome-BGCMuSo performed well in capturing the variability of the AGB, with an R^2 of 0.69 and a RMSE of 31.88 g C m⁻² under the S_{NON-GRAZING} scenario and with an R^2 of 0.65 and a RMSE of 29.42 g C m⁻² under the S_{GRAZING} scenario, when compared with the AGB estimated using a RF algorithm (Fig. 5c1, c2). The regional average AGB was estimated using a RF algorithm at 36.21 g C m⁻². In contrast, the model-simulated AGB under S_{NON-GRAZING} scenario was higher, with an average of 42.35 g C m⁻², whereas the S_{GRAZING} scenario yielded a simulated AGB of 37.22 g C m⁻². Fig. 5d illustrates the observed negative impacts of increasing grazing pressure on vegetation, as well as the ability of the model to simulate these effects. Specifically, the observed AGB decreased as grazing intensity increased from 0 to 1 yak/ha. However, when only leaf carbon (LeafC) and soft stem carbon (SoftstemC) were considered, the model did not reflect this observed trend, instead showing a slight increase under the same grazing conditions (Fig. 5d1). In contrast, the model prediction including LeafC, SoftstemC and STDB more closely matched the

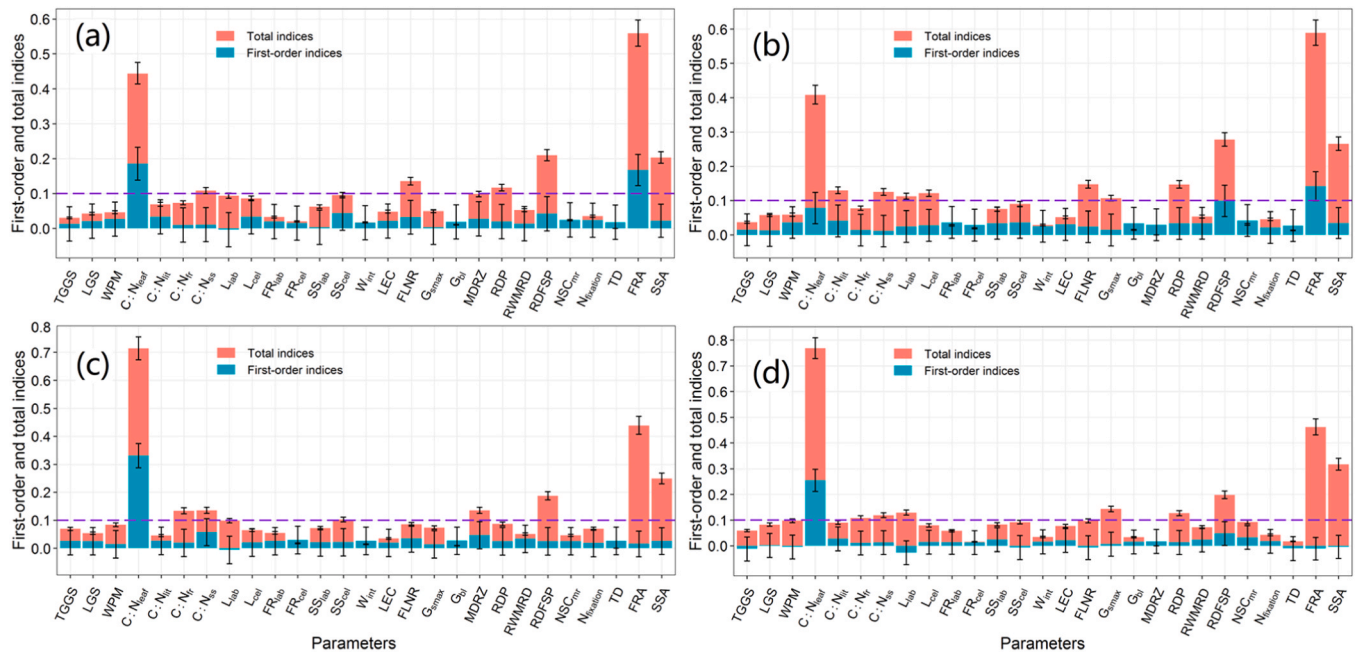


Fig. 3. Sobol sensitivity indices for 27 ecophysiological parameters from Biome-BGCMuSo model at alpine meadow and alpine steppe station. Total-order sensitivity indices (pink), first-order sensitivity indices (blue), and 95 % confidence intervals for aboveground biomass (AGB) of alpine meadow (a), AGB of alpine steppe (b), soil organic carbon (SOC) of alpine meadow (c), and SOC of alpine steppe (d) were shown, respectively. The purple dotted line designates highly sensitive parameters defined using a threshold value of 0.1. [Table S5](#) presents the description and abbreviation of each parameter.

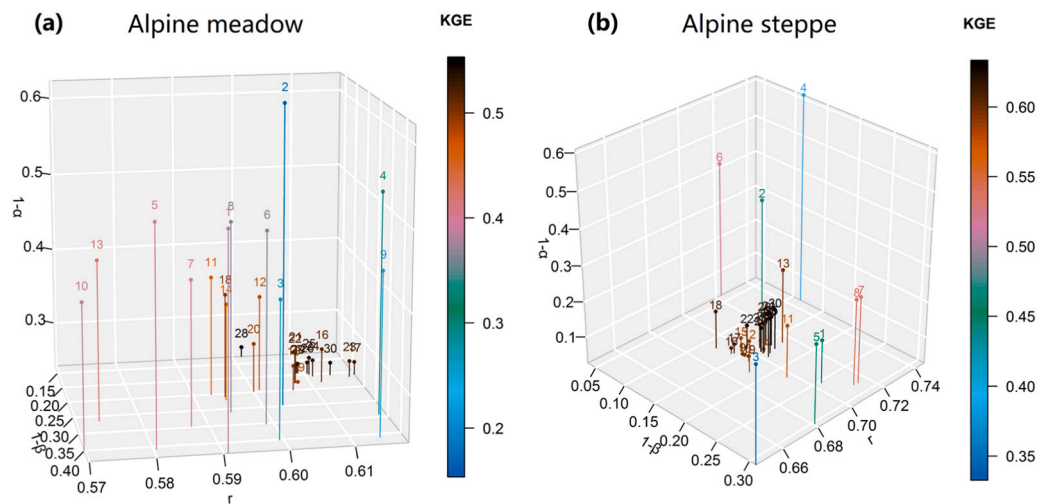


Fig. 4. Changes of optimal results for alpine meadow (a) and alpine steppe (b) in terms of the correlation coefficient (r), the standard deviation ratio (α), and the mean value ratio (k) between the observed and simulated values based on particle swarm optimization (PSO) algorithm.

observed trend (Fig. 5d2). As for increments of 1–2 yak/ha and 2–3 yak/ha, the negative effect of increased grazing intensity on AGB was accurately depicted in a negative direction, reproducing the observed patterns of AGB reduction (Fig. 5d1, d2).

3.2. Grazing effects

Fig. 6 shows the differences in the time series of vegetation carbon (VEGC), litter carbon (LTRC), and top layer (0–30 cm) soil organic carbon (SOC30) for the non-grazing and S_{GRAZING} scenario, representing situations with and without grazing. During 1979–2018, the model simulations revealed grazing alone reduced VEGC, LTRC, and SOC30 by 27.8 g C m^{-2} , 17.0 g C m^{-2} , and 270.1 g C m^{-2} , respectively, indicating that grazing reduced the carbon storage capacity of Qinghai-Tibetan

Plateau. Increasing grazing intensity would also increase the grazing effects on carbon stocks, AGB, and NPP, but there were differences between summer and winter grazing effects on AGB, NPP, and SOC based on model simulations. At a grazing intensity of 10 LSU ha^{-1} or more, summer grazing reduced the VEGC by an average of 226.5 g C m^{-2} , while winter grazing reduced the VEGC by an average of 147.6 g C m^{-2} (Fig. 7a). For LTRC and SOC30, the effect of grazing was more pronounced in the summer pastures than in the winter pastures, especially for SOC30. For instance, we found that summer grazing reduced LTRC by an average of 171.4 g C m^{-2} and SOC30 by an average of $4756.7 \text{ g C m}^{-2}$, whereas winter grazing reduced LTRC by an average of 43.0 g C m^{-2} and SOC30 by an average of 535.2 g C m^{-2} (Fig. 7b, c). There was significant spatial heterogeneity in the effects of grazing on carbon stocks, AGB, and NPP (Figs. 8, S8), as indicated by model simulations. In

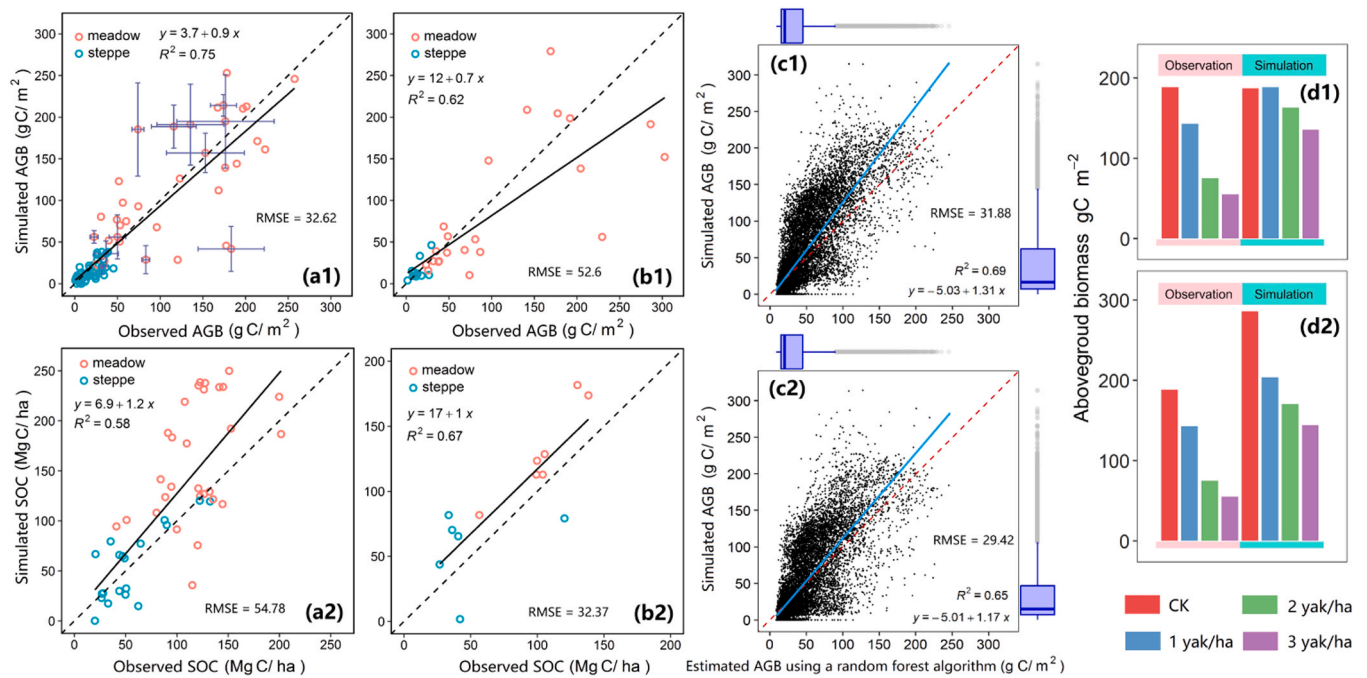


Fig. 5. Scatterplots of the simulated aboveground biomass (AGB) and soil organic carbon (SOC) against field measurements during calibration (a1, a2) and validation (b1, b2). Scatterplot of the simulated AGB and estimated AGB using a random forest (RF) algorithm under $S_{NON-GRAZING}$ scenario (c1) and $S_{GRAZING}$ scenario (c2). Comparison of AGB under different grazing intensities in 2008, contrasting observed data with simulated values for two scenarios: (d1) including leaf carbon (LeafC) and soft stem carbon (SoftstemC), and (d2) including LeafC, SoftstemC, and standing dead biomass (STDB) carbon.

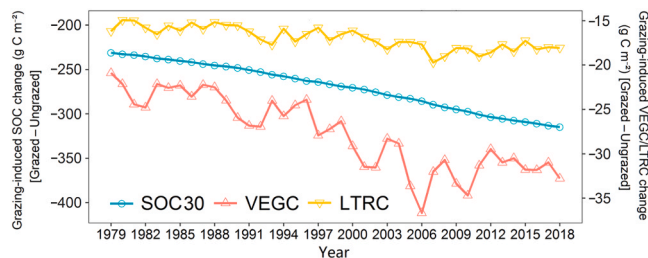


Fig. 6. Differences in vegetation carbon (VEGC), litter carbon (LTRC), and top layer (0–30 cm) soil organic carbon (SOC30) (calculated as grazing–non-grazing) simulated under $S_{NON-GRAZING}$ scenario and $S_{GRAZING}$ scenario in grasslands of the Qinghai-Tibetan Plateau from 1979 to 2018 using Biome-BGCMuSo model.

the Qilian Mountain area of northeastern Qinghai, the Sanjiangyuan area of eastern Qinghai-Tibetan Plateau, the border area between Qinghai and Sichuan, and southeastern Tibet, grazing could reduce total

soil organic carbon (TOTSOC) by more than 1000 g C m^{-2} (Fig. S8a) and VEGC by more than 130.0 g C m^{-2} (Fig. 8b), demonstrating the significant effect of grazing intensity on carbon stocks. In northern Tibet and Qinghai, where grazing intensity was low, grazing had little effect on carbon stocks, AGB, and NPP.

3.3. Spatio-temporal changes of carbon dynamics

3.3.1. Temporal variation in carbon dynamics

Under the simulated $S_{GRAZING}$ scenario, the model simulations revealed a notable temporal variation in carbon dynamics in the grasslands of Qinghai-Tibetan Plateau. In the past 40 years, total carbon (TOTC, defined as the sum of SOC, VEGC, and LTRC), total soil organic carbon (TOTSOC), total carbon above 30 cm soil depth (TOTC30), top layer (0–30 cm) soil organic carbon (SOC30), litter carbon (LTRC), and vegetation carbon (VEGC) showed a significant increasing trend (Figs. 9, S9), with an increase rate of $6.5 \text{ g C m}^{-2} \text{ a}^{-1}$, $4.5 \text{ g C m}^{-2} \text{ a}^{-1}$, $5.2 \text{ g C m}^{-2} \text{ a}^{-1}$, $3.2 \text{ g C m}^{-2} \text{ a}^{-1}$, $0.8 \text{ g C m}^{-2} \text{ a}^{-1}$, and $0.5 \text{ g C m}^{-2} \text{ a}^{-1}$, respectively. TOTSOC and SOC30 variations remained steady between 1979 and 2002, but subsequently increased significantly. Since SOC is the

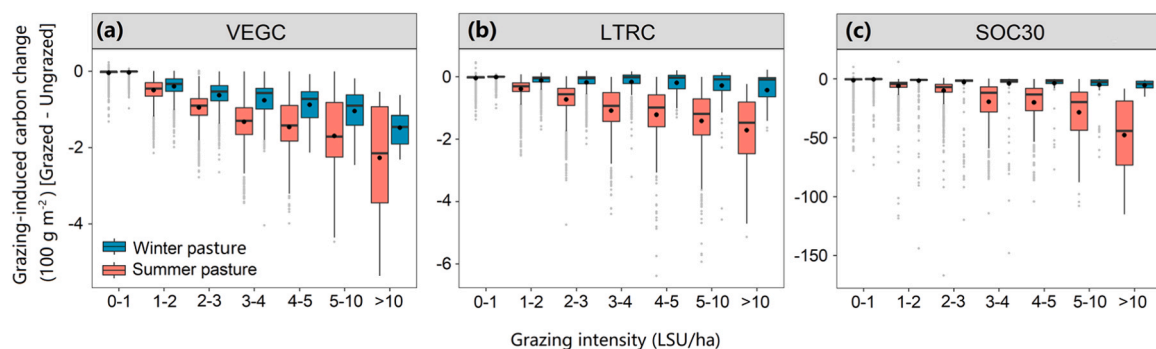


Fig. 7. Under different grazing intensity and seasonal pasture, differences in vegetation carbon (VEGC) (a), litter carbon (LTRC) (b), and top layer (0–30 cm) soil organic carbon (SOC30) (c) between $S_{NON-GRAZING}$ scenario and $S_{GRAZING}$ scenario (calculated as grazing–non-grazing).

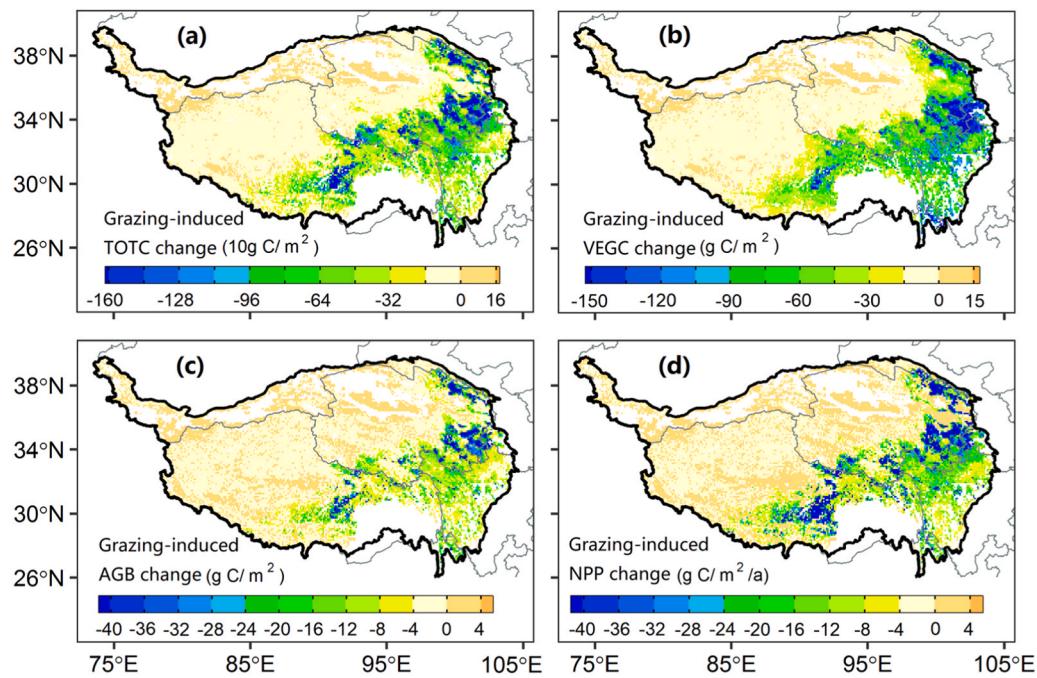


Fig. 8. Spatial differences of total carbon (TOTC, defined as the sum of soil organic carbon, vegetation carbon, and litter carbon) (a), vegetation carbon (VEGC) (b), aboveground biomass (AGB) (c), and net primary productivity (NPP) (d) simulated under $S_{\text{NON-GRAZING}}$ scenario and S_{GRAZING} scenario (calculated as grazing–non-grazing) using Biome-BGCMuSo model.

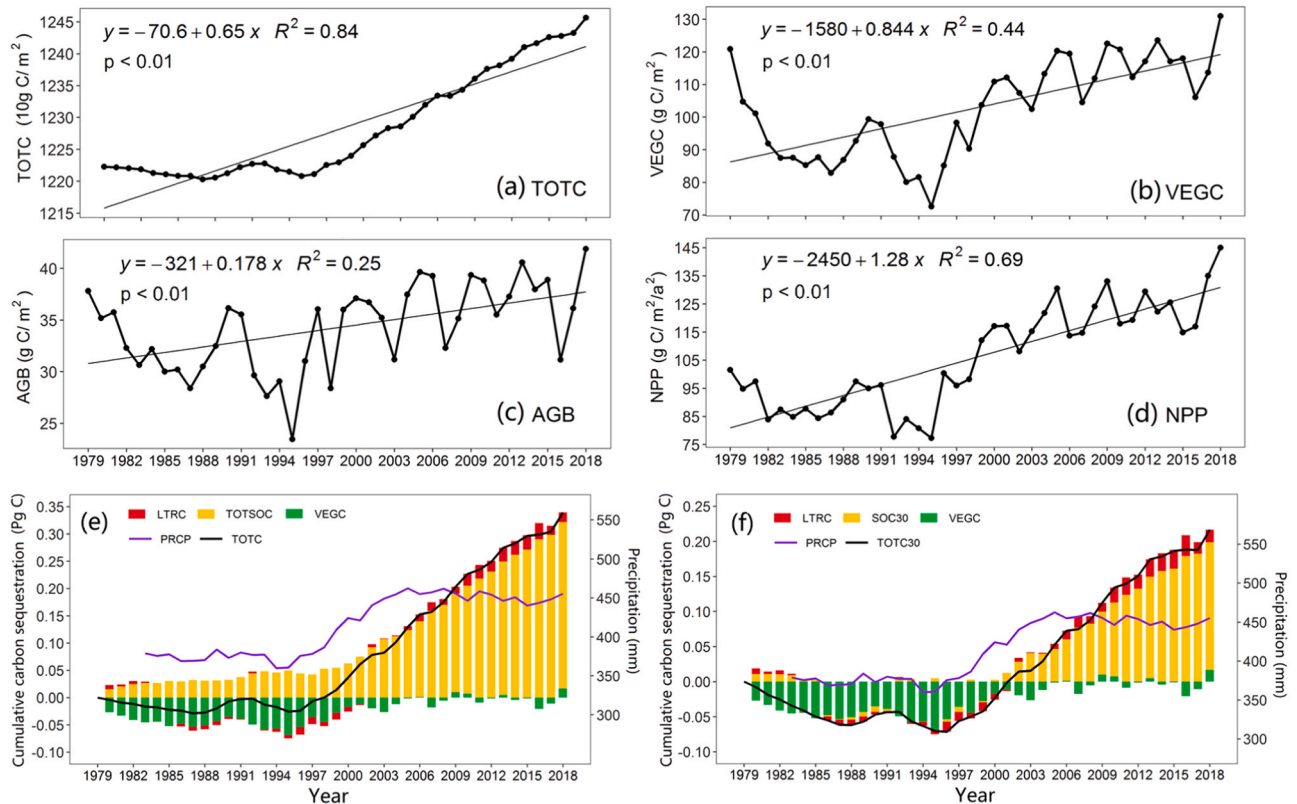


Fig. 9. Interannual variations in total carbon (TOTC, defined as the sum of soil organic carbon, vegetation carbon, and litter carbon) (a), vegetation carbon (VEGC) (b), aboveground biomass (AGB) (c), and net primary productivity (NPP) (d) in the grasslands of the Qinghai-Tibetan Plateau during 1979–2018, based on Biome-BGCMuSo model simulations. Cumulative carbon sequestration since 197 in different carbon pools: total carbon (e), and total carbon above 30 cm soil depth (f). Precipitation (PRCP) value is a five-year moving average.

main component of TOTC, the trend of TOTC and TOTC30 is similar to that of TOTSOC and SOC30. The change in LTRC tended to follow the same pattern as that of VEGC, with the lowest values of 72.5 g C m^{-2} in 1995 and 80.3 g C m^{-2} in 1996 for VEGC and LTRC, respectively. In addition, AGB and NPP increased significantly by $0.2 \text{ g C m}^{-2} \text{ a}^{-1}$ and $1.3 \text{ g C m}^{-2} \text{ a}^{-2}$ respectively. The lowest recorded values were 23.5 g C m^{-2} and $77.3 \text{ g C m}^{-2} \text{ a}^{-1}$ in 1995.

The model simulations indicated that the grasslands of the Qinghai-Tibetan Plateau acted as a carbon sink with an annual carbon sequestration rate of 8.72 Tg C a^{-1} ($1 \text{ Tg} = 10^{12} \text{ g}$), which corresponds to a sequestration of 0.34 Pg C ($1 \text{ Pg} = 10^{15} \text{ g}$) over the past 40 years (Fig. 9e). Specifically, grasslands of the Qinghai-Tibetan Plateau acted as a strong carbon sink with a sequestration of 0.26 Pg C and a carbon sequestration of $16.31 \text{ Tg C a}^{-1}$ between 2002 and 2018. The increase in

TOTSOC since 2002 has led to an increase in TOTC (Fig. 9e). TOTC30 sequestration was 0.22 Pg C over the past 40 years, indicating that 77.5 % of the carbon sequestration occurred above the soil depth of 30 cm (Fig. 9f). Qinghai-Tibetan Plateau is primarily composed of the Tibet and Qinghai, both of which act as carbon sinks. Over the past 40 years, the amount of carbon sequestered in Qinghai was 0.12 Pg C (Fig. S10c), accounting for 35.3 % of the total carbon sequestration of the Qinghai-Tibetan Plateau, while the amount of carbon sequestered in Tibet was 0.16 Pg C (Fig. S10a), accounting for 47.1 % of the total carbon sequestration of the Qinghai-Tibetan Plateau. It is noteworthy that the temporal patterns of cumulative carbon sequestration, especially for TOTC and SOC, were closely related to increasing annual precipitation in grasslands of the Qinghai-Tibetan Plateau (Fig. S10). In the past 40 years, the climate on the Qinghai-Tibetan Plateau showed a

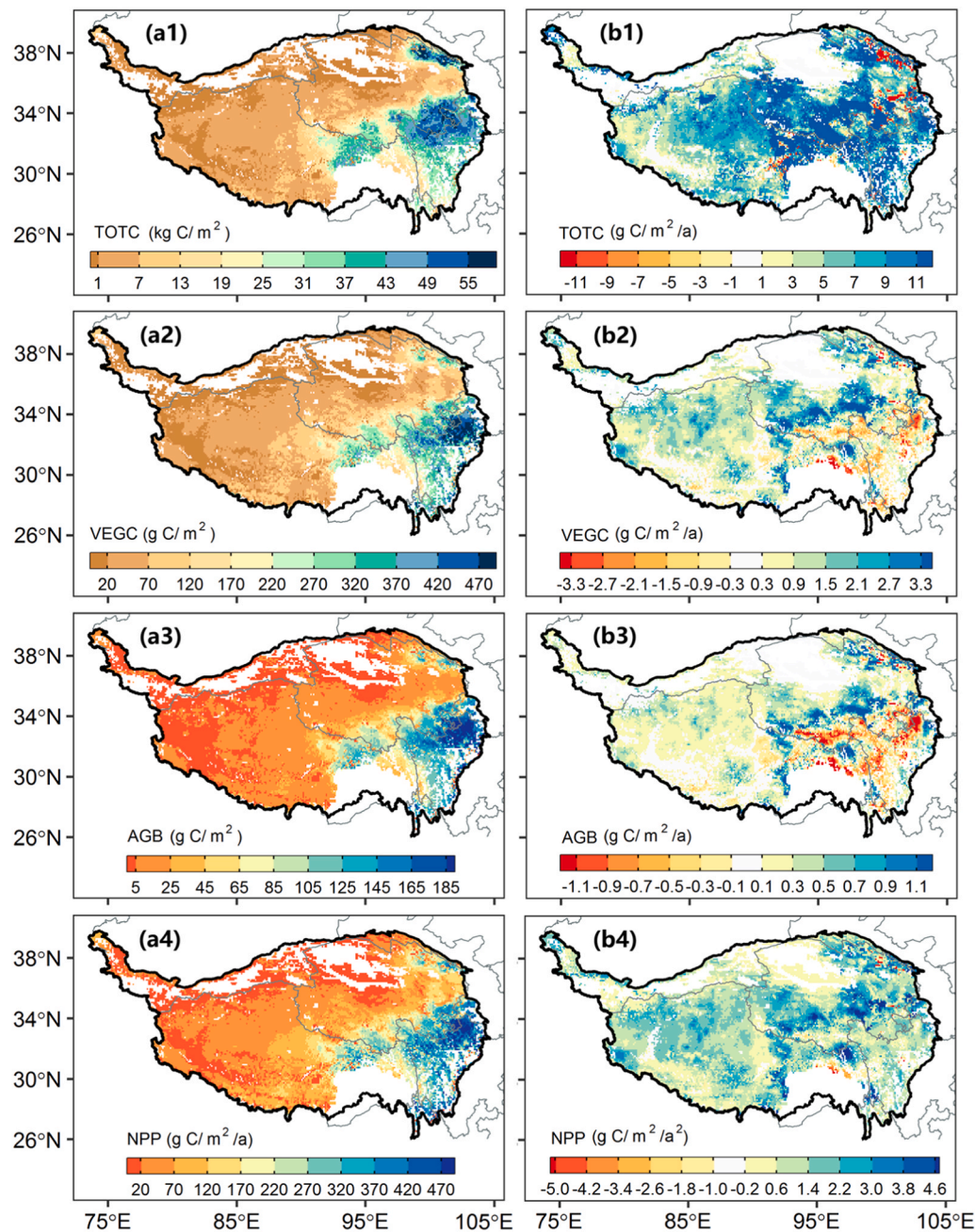


Fig. 10. Distribution of the simulated total carbon (TOTC, defined as the sum of soil organic carbon, vegetation carbon, and litter carbon) (a1), vegetation carbon (VEGC) (a2), aboveground biomass (AGB) (a3), and net primary productivity (NPP) (a4) in grasslands of the Qinghai-Tibetan Plateau during 1979–2018 using the Biome-BGCMuSo model. Change rate of TOTC (b1), VEGC (b2), AGB (b3), and NPP (b4) in grasslands of the Qinghai-Tibetan Plateau during 1979–2018.

“warm-wet” trend, with increases in temperature and precipitation by $0.4\text{ }^{\circ}\text{C}/10\text{a}$ and 3.31 mm a^{-1} , respectively (Fig. S11).

3.3.2. Spatial variation in carbon dynamics

In grasslands of the Qinghai-Tibetan Plateau, the regional average TOTC, TOTSOC, TOTC30, SOC30, LTRC, VEGC, AGB, and NPP were $12284.4\text{ g C m}^{-2}$ (17.01 Pg C), $12087.7\text{ g C m}^{-2}$ (16.75 Pg C), 6054.3 g C m^{-2} (8.70 Pg C), 5857.6 g C m^{-2} (8.43 Pg C), 94.0 g C m^{-2} (0.13 Pg C), 102.7 g C m^{-2} (0.13 Pg C), 34.3 g C m^{-2} (0.044 Pg C), and $105.9\text{ g C m}^{-2}\text{ a}^{-1}$ (0.14 Pg C a^{-1}) under the simulated S_{GRAZING} scenario, respectively. The grassland carbon stocks on the Qinghai-Tibetan Plateau decreased from southeast to northwest in accordance with regional precipitation patterns, as indicated by model simulations (Figs. 10a1, a2, S12a1-a4, S13). Low values ($< 70\text{ g C m}^{-2}$) of VEGC were found mainly in the Qaidam Basin of Qinghai, southern Xinjiang and northern Tibet. Grassland VEGC was highest in Gannan Prefecture of Gansu, Sichuan, and the border region between Qinghai and Sichuan, exceeding 370.0 g C m^{-2} . The spatial distribution patterns of TOTSOC, SOC30, LTRC, AGB, and NPP were all similar to that of VEGC. TOTSOC (SOC30) was the main component of TOTC (TOTC30), which also creates two high-value distribution areas in the east and northeast of the Qinghai-Tibetan Plateau.

The temporal trends in carbon stocks, AGB, and NPP, as simulated by the Biome-BGCMuSo model with spatio-temporal variation of climate in Qinghai-Tibetan Plateau over the past 40 years are shown in Figs. 10b1-b4, S12b1-b4. The majority of the grasslands on the Qinghai-Tibetan Plateau showed strong carbon sink trends, especially in southern Qinghai and eastern Tibet, where the TOTC increased by more than $10.0\text{ g C m}^{-2}\text{ a}^{-1}$ (Fig. 10b1). The southern part of Qinghai played an increasingly important role as a carbon sink during 2000–2018, with TOTC increasing by more than $30.0\text{ g C m}^{-2}\text{ a}^{-1}$ (Fig. S14). In contrast, carbon sources were scattered throughout the southeastern and eastern parts of Qinghai and central Tibet, where TOTSOC and SOC30 declined at different rates (Figs. 10b1, S12b1-b3). The distribution of carbon source areas was consistent with the spatial distribution of high grazing intensity areas (Fig. S4), which accounted for less than 3.0 % of the study area. The strong correlation between the changes in TOTSOC density and the changes in TOTC density indicated that SOC was the main component controlling the regional carbon dynamics in response to changes in climate and CO_2 (Figs. 10b1, S12a2). VEGC increased in 75.5 % of the study area during 1979–2018. The regions with decreasing VEGC were mainly located at the borders between Tibet and Qinghai, and Qinghai and Sichuan (Fig. 10b2). During 2000–2018, VEGC showed an increasing trend in 45.5 % of the study area, mainly in northern Sichuan and southern Qinghai, while VEGC showed a decreasing trend in 54.5 % of the study area, mainly in northern Tibet and western Qinghai (Fig. S14f). The spatial variation pattern of LTRC, AGB and NPP was similar to that of VEGC.

4. Discussion

4.1. Comparisons with previous studies and potential implications

The model predicted that the average AGB for grasslands of the Qinghai-Tibetan Plateau was 34.3 g C m^{-2} during 1979–2018 (with a minimum of 23.5 g C m^{-2} and a maximum of 41.9 g C m^{-2}). In comparison to studies utilizing regression models or random forest (RF) methods based on remote sensing, our simulated AGB is in consistent with those reported by Yang et al. (2009b) at 30.9 g C m^{-2} , Xia et al. (2018) at 35.3 g C m^{-2} , Zeng et al. (2019) at 36.2 g C m^{-2} , and Gao et al. (2020) at 26.8 g C m^{-2} . The cited AGB estimations fall within the range of our simulated AGB minimum and maximum values. The diverse independent AGB values derived from different research methodologies demonstrate the robustness and reliability of the simulation of AGB. The average simulated NPP for the period 1979–2018 was 105.9 g C m^{-2} , which is slightly lower than the findings of the remote sensing-based

light use efficiency (LUE) model. These studies have reported average NPP values of 120.8 g C m^{-2} during 1982–2009 (Zhang et al., 2014), 125.0 g C m^{-2} during 1982–1999 (Piao et al., 2006), and 121.9 g C m^{-2} during 2000–2012 (Xu et al., 2016). Simulations of our model showed a high degree of temporal similarity with the Community Land Model 5.0 (CLM 5.0) model regarding the dynamics of NPPs, despite the differences in the magnitude of the values (Li et al., 2024). This difference in magnitude was largely due to the fact that the CLM5.0 model simulated the entire Qinghai-Tibetan Plateau (encompassing all ecosystems), not only alpine grasslands, providing a higher overall estimate of NPP than our simulations. Nevertheless, certain process-based models were found to overestimate AGB and NPP when compared with our simulation results (Tan et al., 2010; Yi et al., 2014; Zhuang et al., 2010). This discrepancy may be attributed to differences between the parameterizations, or the specific assumptions that are embodied in the frameworks of each model. Aside from the magnitude of the simulations, we have also focused on their spatial distribution. Our findings indicate a gradual increase in AGB and NPP from the southeast to the northwest of the Qinghai-Tibetan Plateau in response to the precipitation gradient (Figs. 10a3, a4, S13). The findings of this study are in accordance with those of previous studies (Zeng et al., 2019; Zhang et al., 2014), which also reported similar spatial gradients. This spatial consistency underscores the importance of environmental gradients in shaping grassland vegetation and carbon dynamics. As for SOC, given the intricate subsurface processes and the multitude of biotic and abiotic interactions that govern SOC input and turnover, simulating the dynamics of SOC is a complex process, with large uncertainties (Huntzinger et al., 2020; Moyano et al., 2013). In the framework of the Multi-Scale Synthesis and Terrestrial Model Intercomparison Project (MsTMIP), 11 ecosystem models were used to explore the dynamics of SOC on the Qinghai-Tibetan Plateau (Ding et al., 2019). The results demonstrated a considerable degree of variability among the models, with SOC estimates ranging from less than 3 Pg C to over 40 Pg C on the Qinghai-Tibetan Plateau. The wide range of outcomes highlights the difficulties in accurately modeling SOC processes, which are influenced by the differences in decomposition rate and its sensitivity to temperature, soil moisture, and carbon-nitrogen interactions in different models (Ito et al., 2020). The SOC simulated in this study (16.75 Pg C) falls within the range of the aforementioned MsTMIP simulations on the Qinghai-Tibetan Plateau. Nevertheless, our SOC estimates are higher than those reported in two previous studies. For example, Zhuang et al. (2010) estimated SOC at $11420.0\text{ g C m}^{-2}$, while Tan et al. (2010) estimated SOC at 8600.0 g C m^{-2} , both of which are lower than our findings ($12284.0\text{ g C m}^{-2}$). For the soil module of Biome-BGCMuSo, the converging cascade method (Thornton, 1998) was used to simulate decomposition, carbon and nitrogen turnover, and soil CO_2 efflux. Due to the lower temperature characteristics of the Qinghai-Tibetan Plateau grasslands, the decomposition process in the Biome-BGCMuSo model may not reach equilibrium as quickly as in warmer climates. This slower decomposition rate, which is evidenced by the decomposition effect of Biome-BGCMuSo is less sensitive to temperature changes between $0\text{ }^{\circ}\text{C}$ and $10\text{ }^{\circ}\text{C}$ than between $10\text{ }^{\circ}\text{C}$ and $20\text{ }^{\circ}\text{C}$ (Hidy et al., 2022). Consequently, there is a possibility that excessive SOC accumulation may occur during the spin-up simulations, reflecting the prolonged time needed for the system to reach equilibrium under cooler conditions.

In the alpine grasslands, where decomposition is typically limited by low temperatures, increased precipitation enhances plant-derived carbon input. This increase in carbon input often outweighs the concurrent rise in carbon loss due to warming-induced acceleration of decomposition (Wei et al., 2021), thereby enhancing the carbon sink capacity of alpine grasslands (Wang et al., 2023). In this study, the grasslands of Qinghai-Tibetan Plateau acted as a carbon sink of 8.72 Tg C a^{-1} over 1979–2018, and of 16.31 Tg C a^{-1} over 2000–2018, as simulated by the model (Fig. 9e). Based on eddy covariance flux measurements, field observations, and remote sensing measurements, it is evident that the grasslands of the Qinghai-Tibetan Plateau act as a carbon sink (Piao

et al., 2019; Wang et al., 2023). For example, in the eastern regions of the Qinghai-Tibetan Plateau, Wang et al. (2021) found that the alpine grasslands act as a strong carbon sink of $200\text{--}300\text{ g C m}^{-2}\text{ a}^{-1}$ by using eddy covariance flux measurements conducted around 2010. Based on gridded net ecosystem exchange (NEE) data generated by upscaling flux observations to the regional scale of the Qinghai-Tibetan Plateau grasslands, annual NEE averaged $-52.11 \pm 14.50\text{ Tg C a}^{-1}$, with a range from -26.39 to $-79.19\text{ Tg C a}^{-1}$ over the period from 1982 to 2018 (Wang et al., 2023). The estimated value is considerably higher than that predicted by our simulations. The findings of this study indicate that TOTSOC is the most dominant component of TOTC, with VEGC representing a relatively minor component. Consequently, it can be posited that carbon sinks will eventually lead to an increase in SOC. This inference is supported by multiple studies revealing that the rate of increase of surface SOC (top 30 cm) was $28.0\text{ g C m}^{-2}\text{ a}^{-1}$ during 2000–2010 and $4.7\text{ g C m}^{-2}\text{ a}^{-1}$ during 2002–2011, based on soil surveys (Chen et al., 2017; Ding et al., 2017).

4.2. Effect of grazing on carbon stocks

The high root/shoot ratio characteristic of alpine grasslands (e.g., reaching up to 23.82 in alpine meadows) results in a greater proportion of biomass being allocated below ground (Hafner et al., 2012; Zhang et al., 2025). Meanwhile, the total flux of root exudate was estimated to be highest in grassland ecosystems due to the high densities of absorptive root biomass (Chari et al., 2024). This significant belowground investment directly enhances soil organic carbon input and supports SOC stocks. Consequently, grassland SOC on the Qinghai-Tibetan Plateau is significantly and positively correlated with AGB and belowground biomass, so when AGB and root biomass are reduced by grazing, SOC accumulation is inhibited and the soil may become a carbon source rather than a carbon sink as grazing intensity increases (Sun et al., 2011). In this study, the model suggested that grazing disturbance had significant effects on carbon stocks, AGB, and NPP. The model revealed that grazing generally reduced AGB, VEGC (including root biomass), and SOC in grasslands of the Qinghai-Tibetan Plateau; however, some areas with grazing intensity less than 1.0 LSU ha^{-1} showed over-compensation that stimulates vegetation growth (Fig. 7a). During a 5-year grazing control experiment on the Qinghai-Tibetan Plateau, grazing significantly reduced SOC by 32.9 % (Ma et al., 2016). A meta-analysis of 136 paired observations from 55 publications has quantified the impact of livestock grazing on SOC stocks and AGB in grasslands of the Qinghai-Tibetan Plateau, revealing a significant decrease in SOC stock (11.9 %) and AGB (51.6 %) following grazing activities (Ma et al., 2022). These findings serve to highlight the negative influence of livestock grazing on both SOC and above-ground biomass, thereby emphasizing the necessity for the implementation of sustainable grazing management practices.

In this study, the model suggested that the effect of grazing on VEGC, LTRC and SOC30 increased significantly with increasing grazing intensity in both summer and winter pastures, and the effect of summer grazing on VEGC, LTRC and SOC30 was stronger than that of winter grazing at the same grazing intensity, especially on SOC30 (Fig. 7c). These results are consistent with those of Chang et al. (2014), who found that summer grazing resulted in greater soil carbon loss than winter grazing at the same grazing intensity, suggesting that summer pastures could be converted to winter pastures to increase soil carbon sinks. Two factors account for the difference in the effect of summer grazing on carbon dynamics compared to winter grazing. First, due to physiological adaptations of livestock and less accessible or palatable STDB for livestock, the feed intake of livestock such as Tibetan sheep is inherently lower in winter than in summer (Guo, 2020), resulting in less detrimental effects on STDB compared to AGB during the summer months. Second, summer grazing occurs during the vigorous growth period of plants, resulting in reduced AGB (i.e., leaves and soft stems). This reduction in AGB reduces net photosynthetic carbon assimilation (Wei

et al., 2020), resulting in less carbon available for allocation to below-ground biomass, which in turn directly reduces SOC inputs. In contrast, winter grazing generally affects the STDB and rarely affects the assimilation products allocated belowground during the growing season. This is consistent with previous findings that grassland SOC on the Qinghai-Tibetan Plateau had a strong positive correlation with both AGB and belowground biomass (Sun et al., 2011). Under fencing to exclude grazing, summer pastures with high grazing intensity have a higher carbon sink potential than winter pastures with the same intensity (Fig. 7a-c). Therefore, it is the best option to impose grazing exclusion on summer pastures with high grazing intensity in order to restore the ecological environment. Moreover, the widening difference in simulated carbon stocks between S_{GRAZING} scenario and $S_{\text{NON-GRAZING}}$ scenario indicated that CO_2 enrichment and climate change progressively amplified the carbon sequestration benefits of grazing exclusion (Figs. 6, 7). By enhancing the positive effects of CO_2 fertilization and the warming and wetting climate, grazing exclusion becomes increasingly effective over time under the same grazing intensity.

4.3. Uncertainties and future prospects

To constrain the Biome-BGCMuSo model, we used spatially distributed field-observed data, which mainly represented carbon dynamics under different environmental gradients. Although this approach was necessary due to the lack of long-term time series carbon stock data for ungrazed alpine grasslands, it introduced some uncertainty (Zhu et al., 2022). Consequently, caution is advised when interpreting the mechanisms behind simulated temporal changes (e.g., long-term carbon shifts) at specific locations, as inferred from the model. These uncertainties highlight the need for future long-term grazing experiments (covering a gradient of grazing intensities, and spanning both warm- and cold-season pastures) to validate the simulated temporal trends and refine our understanding of the mechanisms driving carbon dynamics under varying grazing pressures. Despite this limitation, Biome-BGCMuSo model remains a powerful tool for simulating alpine grassland carbon dynamics. (1) The model was driven by daily time-series meteorological data (i.e., the China Meteorological Forcing Dataset) from 1979 to 2018, which enabled dynamic interactions between climate and vegetation; (2) the process-based ecosystem modeling framework of the Biome-BGCMuSo has been validated across diverse grassland ecosystems worldwide, effectively capturing key mechanisms of carbon cycling in grasslands over time (Han et al., 2016; Hidy et al., 2022; Huang et al., 2022; Sándor et al., 2016). Therefore, Biome-BGCMuSo model remains a valuable tool for predicting regional carbon dynamics under grazing pressure in alpine grasslands of the Qinghai-Tibetan Plateau and for informing future management practices.

In this study, only elevation data were used for the division of summer pastures and winter pastures, which is insufficient to accurately describe the actual distribution of seasonal pastures (Zhan et al., 2023). This may be the reason for the observed decrease in the R^2 from the $S_{\text{NON-GRAZING}}$ scenario to the S_{GRAZING} scenario in the simulated AGB (Fig. 5c1, c2). Future research could incorporate population distribution, the boundaries of the National Nature Reserve, and remote sensing data in order to more accurately depict seasonal pastures distribution. In addition, the grazing intensity data for livestock are somewhat uncertain because livestock change dynamically within a year and have an age composition structure, and not all livestock in the statistical yearbook were grazing, some of which were bred in captivity. However, the data in the statistical yearbook do not reflect these changes in livestock over the course of a year. It would be worthwhile to refine the changes in livestock within a year and the age composition structure of livestock in future research. There is no doubt that supplementary feeding is an important part of animal husbandry during the winter months. Supplementary feeding was not considered in this study. Future research could consider and refine the timing of winter supplementary feeding

based on the amount of snowfall. Here, we highlight the potential of integrating remote sensing data to enhance grazing intensity assessments accuracy. For example, Chang et al. (2024) generated annual grazing probability and intensity maps based on Landsat 7/8 and Sentinel-2 images for the Hulun Buir grasslands in China.

The validation results of the grazing module (Fig. 5d) demonstrate the intricate relationship between grazing intensity and AGB, and highlight the importance of accurate model representation, particularly in low grazing intensity. The discrepancy observed at low grazing intensity for LeafC and SoftstemC (Fig. 5d1) may be attributed to three primary factors. First, the model utilizes 0.1° resolution driver data, which may not capture the fine-scale variability observed at individual sites. Second, it is possible that the grazing module does not adequately account for the subtle changes in plant community structure and function that occur with low grazing activities, suggesting a need for improved model sensitivity to these dynamics. Third, the inclusion of STDB as a resource consumed by livestock has a significant impact on the simulation of grazing effects on vegetation. STDB acts as a buffer, partially mitigating the effects of grazing on LeafC and SoftstemC. This substitution of STDB for LeafC and SoftstemC damage has important implications for the model's representation of grazing effects on the ecosystem. It highlights the need to consider the full range of biomass states in grazing models in order to accurately predict the effects of different grazing intensities on vegetation structure and function. Our findings indicate a need for further research into the manner in which models simulate the effects of grazing, particularly at low grazing intensity. In future research, it will also be essential to validate and refine the grazing module of the model based on controlled experiments of winter grazing.

5. Conclusion

In this study, Biome-BGCMuSo, a process-based biogeochemical model, was used to investigate the spatiotemporal variability of grassland carbon dynamics on the Qinghai-Tibetan Plateau under the influence of climatic factors and grazing management. Specifically, the Biome-BGCMuSo model was more capable of simulating seasonal rotational grazing by adding new winter grazing module. To calibrate the model parameters, we combined the global sensitivity analysis method with the PSO algorithm, which resulted in a significant improvement in the model simulations. The carbon dynamics of the grasslands on the Qinghai-Tibetan Plateau were then simulated using the improved Biome-BGCMuSo model. In recent decades, the grasslands of the Qinghai-Tibetan Plateau have acted as a carbon sink, with a notable increase in their ability to store carbon in the soil based on model simulations. This is evidenced by the overall trend of an increase in TOTC, primarily due to the rise in SOC. In both summer and winter pastures, the model revealed that grazing had a negative effect on carbon stocks, AGB, and NPP, and its influence on VEGC, LTRC, and SOC increased significantly with increasing grazing intensity. In contrast, model suggested that summer grazing had stronger effects on VEGC, LTRC, and SOC than winter grazing at the same grazing intensity, especially on SOC. The most efficient way to increase carbon sinks is to prioritize the enclosure of summer pastures with high grazing intensity. Our study not only highlights the critical influence of grazing on the carbon cycle, but also highlights the need for nuanced grazing models that take into account both seasonal variations and their ecological consequences. On the Qinghai-Tibetan Plateau, it would be informative to produce data on the temporal dynamics and spatial distribution of livestock grazing intensity on the summer and winter pastures in the future. The incorporation of more accurate representations of both winter and summer pasture distribution and grazing intensity into process-based models will enable the development of more effective strategies that not only balance the sustainable use of grasslands on the Qinghai-Tibetan Plateau but also maximize their carbon sink potential.

CRedit authorship contribution statement

Lei Zheng: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Yanyan Pei:** Writing – review & editing. **Feng Yang:** Writing – review & editing. **Xiuliang Yuan:** Writing – review & editing, Funding acquisition. **Jie Peng:** Writing – review & editing. **Bingjian Zhu:** Writing – review & editing. **Guoliang Wang:** Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.agee.2025.110056](https://doi.org/10.1016/j.agee.2025.110056).

Data availability

The data generated and analyzed of this study are openly available in the Mendeley Data repository at <https://data.mendeley.com/datasets/5dk5w77kdj/1>.

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